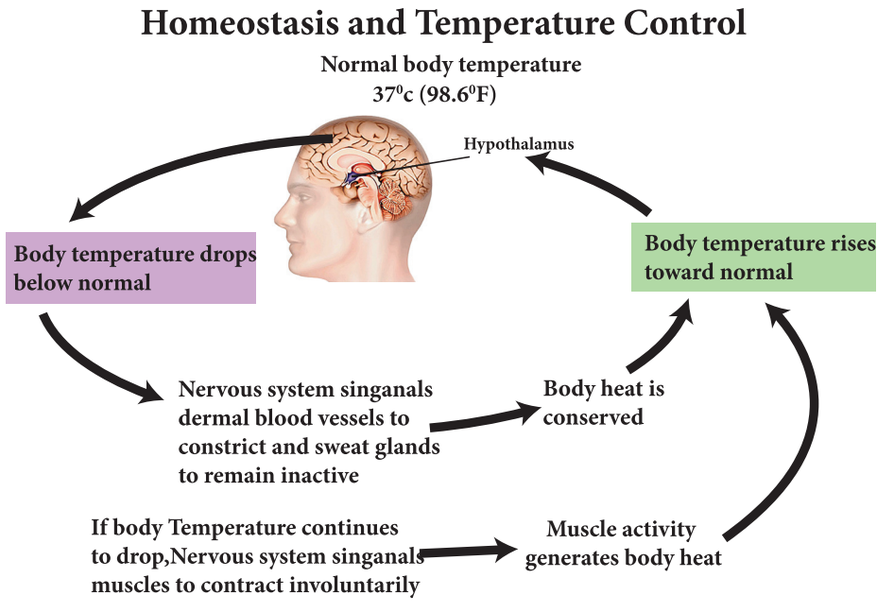


The relationship is regulated by neurological and cardiovascular mechanisms. The nurse applies knowledge of temperature control mechanisms to promote temperature regulation.

TEMPERATURE REGULATION

The hypothalamus located between the cerebral hemispheres, controls the body temperature. The hypothalamus sense main changes in body temperature. The anterior hypothalamus controls the heat production. When the nerve cells in the anterior hypothalamus became heated beyond the set point, impulses are sent out to reduce body temperature. Mechanisms of heat loss include sweating, vasodilatation (widing) of blood vessels and inhibition of heat production. If the posterior hypothalamus senses the body temperature is lower than the set point, heat conservation mechanisms are instituted. Vasoconstriction (narrowing of blood vessels) reduces blood flow to the skin and extremities. Heat production is stimulated through voluntary muscle contraction and muscle shivering.

Factors affecting body temperature

Many factors affect body temperature changes in body temperature. The nurse must be aware of these factors when assessing temperature variables and evaluating deviation from normal.

- **Age:** For new born, the temperature – control mechanism are immature. An infant's temperature may respond drastically to changes in the environment. Temperature regulation is unstable until children reach puberty. Older adults are sensitive to temperature extreme because of deterioration in control mechanisms, reduced sweat gland activity, reduced amounts of subcutaneous fat and reduced metabolism.
- **Exercise:** Muscle activity causes increase metabolism by increasing carbohydrate and fat breakdown. Any

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